```
(.venv) (base) Sergio@SERHIIs-MacBook-Pro homework_buggy % pytest -v
tests/test_iot_network.py
============================= test session starts
==============================
platform darwin -- Python 3.9.12, pytest-8.4.2, pluggy-1.6.0 -- /Users/Sergio/
Documents/QA_Embedded/Gate/Task7/HomeWork/homework_buggy/.venv/bin/
python
cachedir: .pytest_cache
metadata: {'Python': '3.9.12', 'Platform': 'macOS-10.16-x86_64-i386-64bit',
'Packages': {'pytest': '8.4.2', 'pluggy': '1.6.0'}, 'Plugins': {'metadata': '3.1.1',
'html': '4.2.0'}, 'JAVA_HOME': '/Library/Java/JavaVirtualMachines/jdk-21.jdk/
Contents/Home'}
rootdir: /Users/Sergio/Documents/QA_Embedded/Gate/Task7/HomeWork/
homework_buggy
plugins: metadata-3.1.1, html-4.2.0
collected 20 items

tests/test_iot_network.py::test_tc01_states_observable PASSED            [  5%]
tests/test_iot_network.py::test_tc02_no_hang_in_outage PASSED
[ 10%]
tests/test_iot_network.py::test_tc03_backoff_no_storm FAILED             [ 15%]
tests/test_iot_network.py::test_tc04_offline_buffer PASSED               [ 20%]
tests/test_iot_network.py::test_tc05_recovery_flush FAILED               [ 25%]
tests/test_iot_network.py::test_tc06_overflow_counters FAILED            [ 30%]
tests/test_iot_network.py::test_tc07_drop_oldest_order FAILED            [ 35%]
tests/test_iot_network.py::test_tc08_boundary_8 PASSED                   [ 40%]
tests/test_iot_network.py::test_tc09_boundary_9 FAILED                   [ 45%]
tests/test_iot_network.py::test_tc10_no_duplicate_ids FAILED             [ 50%]
tests/test_iot_network.py::test_tc11_sent_consistency FAILED             [ 55%]
tests/test_iot_network.py::test_tc12_heap_stability FAILED               [ 60%]
tests/test_iot_network.py::test_tc13_stress_offline FAILED               [ 65%]
tests/test_iot_network.py::test_tc14_status_fields PASSED                [ 70%]
tests/test_iot_network.py::test_tc15_reset_stats FAILED                  [ 75%]
tests/test_iot_network.py::test_tc16_invalid_commands[status] PASSED
[ 80%]
tests/test_iot_network.py::test_tc16_invalid_commands[STRESS] PASSED
[ 85%]
tests/test_iot_network.py::test_tc16_invalid_commands[FOO] PASSED
[ 90%]
tests/test_iot_network.py::test_tc17_stress_out_of_range[0] PASSED
[ 95%]
tests/test_iot_network.py::test_tc17_stress_out_of_range[501] PASSED
[100%]

================================== FAILURES
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_____________________________ test_tc03_backoff_no_storm
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_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc03_backoff_no_storm(node):
        out = node.cmd("NET DOWN", 20)
        ts = [t for t, l in out if "[NET] retry=" in l]
        assert ts, "no retry lines after NET DOWN"
>       assert len(ts) <= 12, f"{len(ts)} retries in 20s = reconnect storm"
E       AssertionError: 391 retries in 20s = reconnect storm
E       assert 391 <= 12
E        +  where 391 = len([1792854927.549736, 1792854928.049212,
1792854928.0976, 1792854928.1477408, 1792854928.199959,
1792854928.248697, ...])

tests/test_iot_network.py:90: AssertionError
_____________________________ test_tc05_recovery_flush
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc05_recovery_flush(node):
        ids = node.offline_send(5)
        out = node.cmd("NET UP", 8)
>       assert sorted(published(out)) == sorted(ids)
E       AssertionError: assert [6, 6, 7, 7, 8, 8, ...] == [6, 7, 8, 9, 10]
E
E         At index 1 diff: 6 != 7
E         Left contains 5 more items, first extra item: 8
E
E         Full diff:
E           [
E               6,...
E
E         ...Full output truncated (10 lines hidden), use '-vv' to show

tests/test_iot_network.py:109: AssertionError
_____________________________ test_tc06_overflow_counters
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc06_overflow_counters(node):
        node.offline_send(12)
        st = node.status()
>       assert st["buffer"] == CAP and st["dropped"] == 4
E       assert (12 == 8)
```

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tests/test_iot_network.py:117: AssertionError
_____________________________ test_tc07_drop_oldest_order
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc07_drop_oldest_order(node):
        ids = node.offline_send(12)
>       assert published(node.cmd("NET UP", 10)) == ids[-CAP:]
E       AssertionError: assert [31, 31, 32, 32, 33, 33, ...] == [27, 28, 29, 30, 31,
32, ...]
E
E         At index 0 diff: 31 != 27
E         Left contains 16 more items, first extra item: 27
E
E         Full diff:
E           [
E         +     31,...
E
E         ...Full output truncated (24 lines hidden), use '-vv' to show

tests/test_iot_network.py:124: AssertionError
_____________________________________ test_tc09_boundary_9
_____________________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc09_boundary_9(node):
        ids = node.offline_send(9); st = node.status()
>       assert st["buffer"] == 8 and st["dropped"] == 1
E       assert (9 == 8)

tests/test_iot_network.py:136: AssertionError
_____________________________ test_tc10_no_duplicate_ids
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc10_no_duplicate_ids(node):
        node.offline_send(8)
        pub = published(node.cmd("NET UP", 10))
>       assert len(pub) == len(set(pub)), f"duplicate IDs: {pub}"
E       AssertionError: duplicate IDs: [52, 52, 53, 53, 54, 54, 55, 55, 56, 56, 57,
57, 58, 58, 59, 59]
E       assert 16 == 8
E        +  where 16 = len([52, 52, 53, 53, 54, 54, ...])
```

```
E       +  and   8 = len({52, 53, 54, 55, 56, 57, ...})
E       +    where {52, 53, 54, 55, 56, 57, ...} = set([52, 52, 53, 53, 54, 54, ...])

tests/test_iot_network.py:144: AssertionError
_____________________________ test_tc11_sent_consistency
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc11_sent_consistency(node):
        node.offline_send(12); node.cmd("NET UP", 10)
        st = node.status()
>       assert st["sent"] == 12 - st["dropped"] == 8
E       assert 24 == (12 - 0)

tests/test_iot_network.py:152: AssertionError
_____________________________ test_tc12_heap_stability
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc12_heap_stability(node):
        heaps = [node.status()["heap"]]
        for _ in range(10):
            node.cmd("STRESS 100", 6); heaps.append(node.status()["heap"])
        drops = [a - b for a, b in zip(heaps, heaps[1:])]
        print("heap:", heaps, "bytes/msg:", [d / 100 for d in drops])
>       assert not all(d > 0 for d in drops), "heap fell on every iteration"
E       AssertionError: heap fell on every iteration
E       assert not True
E        +  where True = all(<generator object
test_tc12_heap_stability.<locals>.<genexpr> at 0x7fdc9888e820>)

tests/test_iot_network.py:162: AssertionError
---------------------------- Captured stdout call
-----------------------------
heap: [344924, 336924, 328932, 320920, 312920, 304920, 296920, 288920,
280920, 272920, 264920] bytes/msg: [80.0, 79.92, 80.12, 80.0, 80.0, 80.0,
80.0, 80.0, 80.0, 80.0]
_____________________________ test_tc13_stress_offline
_____________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc13_stress_offline(node):
        node.cmd("NET DOWN", 1); node.cmd("STRESS 100", 8); st =
node.status()
```

```
>       assert st["buffer"] <= CAP and st["dropped"] == 100 - CAP
E       assert (100 <= 8)

tests/test_iot_network.py:169: AssertionError
_____________________________________ test_tc15_reset_stats
_____________________________________

node = <test_iot_network.Node object at 0x7fdc98723a30>

    def test_tc15_reset_stats(node):
        node.offline_send(10); node.cmd("RESET STATS", 1); st = node.status()
>       assert (st["sent"], st["dropped"], st["duplicates"],
st["reconnect_attempts"]) == (0, 0, 0, 0)
E       AssertionError: assert (0, 0, 0, 21) == (0, 0, 0, 0)
E
E         At index 3 diff: 21 != 0
E
E         Full diff:
E           (
E               0,
E               0,...
E
E         ...Full output truncated (6 lines hidden), use '-vv' to show

tests/test_iot_network.py:182: AssertionError
============================== short test summary info
==============================
FAILED tests/test_iot_network.py::test_tc03_backoff_no_storm -
AssertionError: 391 retries in 20s = reconnect storm
FAILED tests/test_iot_network.py::test_tc05_recovery_flush - AssertionError:
assert [6, 6, 7, 7, 8, 8, ...] == [6, 7, 8, 9, 10]
FAILED tests/test_iot_network.py::test_tc06_overflow_counters - assert (12 ==
8)
FAILED tests/test_iot_network.py::test_tc07_drop_oldest_order -
AssertionError: assert [31, 31, 32, 32, 33, 33, ...] == [27, 28, 29, 30, 31, ...
FAILED tests/test_iot_network.py::test_tc09_boundary_9 - assert (9 == 8)
FAILED tests/test_iot_network.py::test_tc10_no_duplicate_ids - AssertionError:
duplicate IDs: [52, 52, 53, 53, 54, 54, 55, 55, 56, 56, 57, 5...
FAILED tests/test_iot_network.py::test_tc11_sent_consistency - assert 24 ==
(12 -0)
FAILED tests/test_iot_network.py::test_tc12_heap_stability - AssertionError:
heapfell on every iteration
FAILED tests/test_iot_network.py::test_tc13_stress_offline - assert (100 <= 8)
FAILED tests/test_iot_network.py::test_tc15_reset_stats - AssertionError:
assert (0, 0, 0, 21) == (0, 0, 0, 0)
==================== 10 failed, 10 passed in 367.35s (0:06:07)
====================
```

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